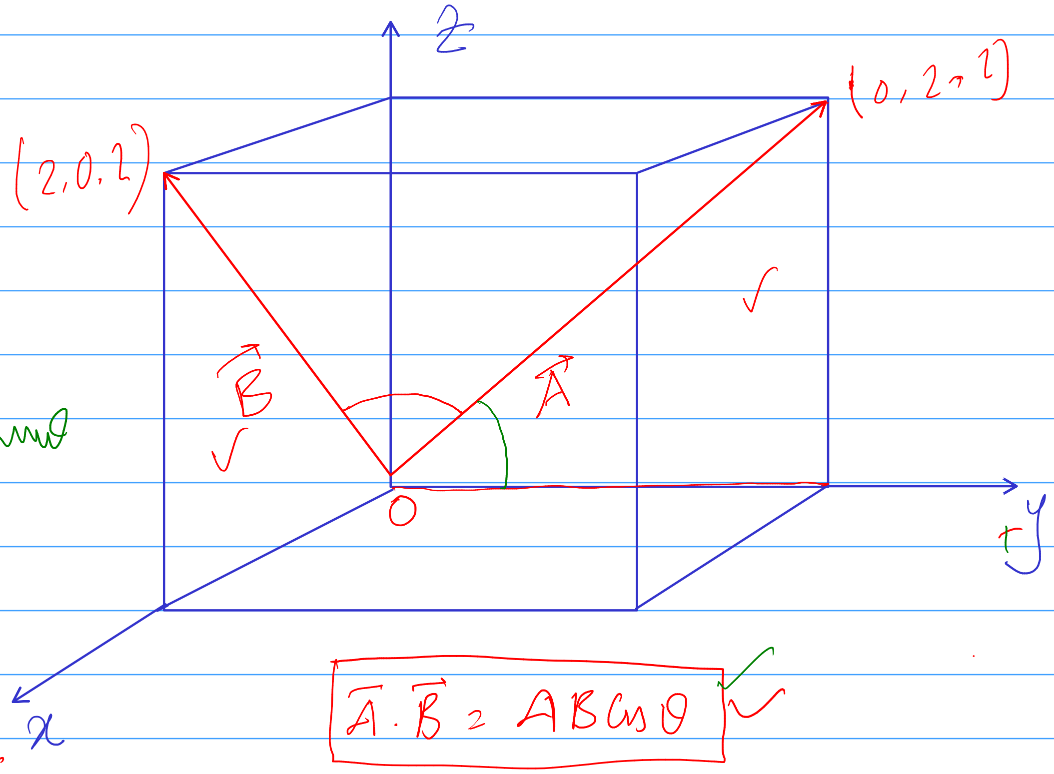
Find the angle between the face diagonals of a cube of edge length 2 m.



$\vec{A} \cdot \hat{j} = |A||\hat{j}| \cos\theta$

$$\boxed{\vec{A} \cdot \vec{B} = AB\cos\theta}$$

$\vec{A} = 0\hat{i} + 2\hat{j} + 2\hat{k} = 2\hat{j} + 2\hat{k}$

$\vec{B} = 2\hat{i} + 0\hat{j} + 2\hat{k} = 2\hat{i} + 2\hat{k}$

$|A| = \sqrt{8} = 2\sqrt{2}$

$|B| = 2\sqrt{2}$

$\vec{A} \cdot \vec{B} = (2\hat{j} + 2\hat{k}) \cdot (2\hat{i} + 2\hat{k}) = 4$

$\Rightarrow \quad \vec{A} \cdot \vec{B} = AB\cos\theta$

$\Rightarrow \quad 4 = 2\sqrt{2} \times 2\sqrt{2} \cos\theta$

$\Rightarrow \quad 4 = 8\cos\theta$

$\Rightarrow \quad \theta = \cos^{-1}(1/2) = 60^\circ$

Consider three vectors, $\vec{a} = 5\hat{\imath} + 4\hat{\jmath} - 6\hat{k}\ m$, $\vec{b} = -2\hat{\imath} + 2\hat{\jmath} + 3\hat{k}\ m$ and $\vec{c} = 4\hat{\imath} + 3\hat{\jmath} + 2\hat{k}\ m$

(i) In unit vector notation find a vector $\vec{r}$ such that, $\vec{r} = \vec{a} - \vec{b} + \vec{c}$.

(ii) Calculate the angle between $\vec{a}$ and $\vec{b}$.

(iii) What is the component of vector $\vec{a}$ along the direction of $\vec{b}$?

(iv) What is the component of $\vec{a}$ perpendicular to the direction of $\vec{b}$, but in the plane of $\vec{a}$ and $\vec{b}$.

(v) Calculate the angle between $\vec{r}$ and positive z-axis.

(vi) Calculate the $\vec{a} \cdot (\vec{b} \times \vec{c})$. What is the unit of this quantity.

Soln:

$$\vec{r} = \vec{a} - \vec{b} + \vec{c}$$

$$\vec{r} = 5\hat{i} + 4\hat{j} - 6\hat{k} - (-2\hat{i} + 2\hat{j} + 3\hat{k}) + 4\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\vec{r} = 5\hat{i} + 4\hat{j} - 6\hat{k} + 2\hat{i} - 2\hat{j} - 3\hat{k} + 4\hat{i} + 3\hat{j} + 2\hat{k}$$

$$\boxed{\vec{r} = 11\hat{i} + 5\hat{j} - 7\hat{k}} \text{ Ans}$$

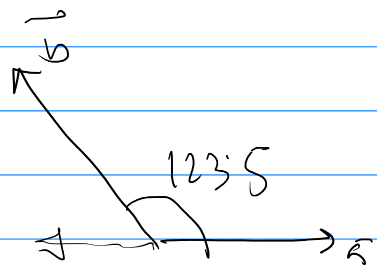
(ii) $\vec{a} \cdot \vec{b} = ab\cos\theta$ $\qquad |\vec{a}| = \sqrt{77}$, $|\vec{b}| = \sqrt{17}$

$$\Rightarrow -10 + 8 - 18 = \sqrt{77}\sqrt{17}\cos\theta$$

$$\Rightarrow -20 = \sqrt{77}\sqrt{17}\cos\theta$$

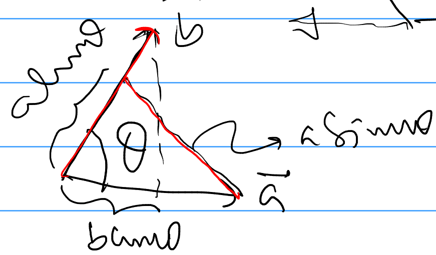
$$\Rightarrow \theta = \cos^{-1}\left(\frac{-20}{\sqrt{77}\sqrt{17}}\right)$$

$$\Rightarrow \boxed{\theta = 123.5^\circ}$$



(iii) $a\cos\theta = \sqrt{77}\cos 123.5^\circ$

$$\boxed{a\cos\theta = -4.8\ m}$$



$\vec{a}\cdot\vec{b} = ab\cos\theta$

$\Rightarrow a\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$

$= -4.8\ m$

(iv) $a \sin\theta = \sqrt{77} \sin 123.5^\circ = 7.3\,m$

$|\vec{a} \times \vec{b}| = |a||b| \sin\theta$

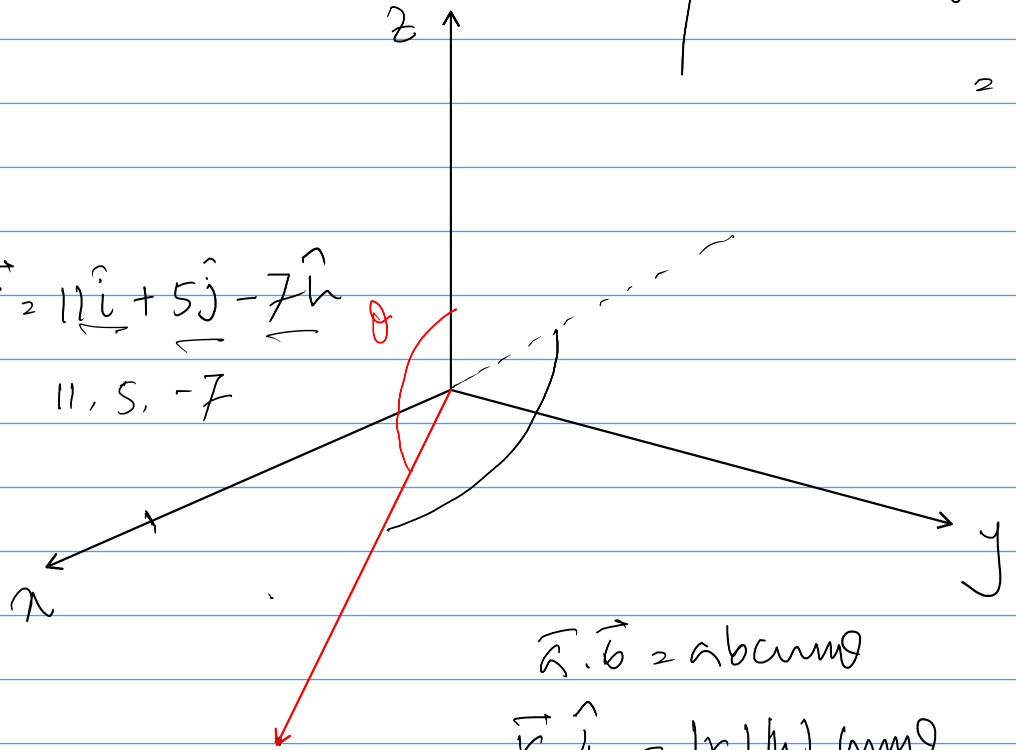
$= a \sin\theta = \frac{\vec{a} \times \vec{b}}{|b|}$

$= +7.3\,m$

(v)

$\vec{r} = 11\hat{i} + 5\hat{j} - 7\hat{k}$

11, 5, -7



$\vec{a} \cdot \vec{b} = ab\cos\theta$

$\vec{r} \cdot \hat{k} = |r||k| \cos\theta$

$= -7 = \sqrt{195} \cdot 1 \cos\theta$

$= \theta = \cos^{-1}\left(\frac{-7}{\sqrt{195}}\right)$

$= \boxed{\theta = 120.1^\circ}$

(vi) $\vec{a} \cdot (\vec{b} \times \vec{c}) =$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & 3 \\ 4 & 3 & 2 \end{vmatrix}$$

(j) (k) (x)

$= \hat{i}(B_y C_z - B_z C_y) + \hat{j}(B_z C_x - B_x C_z) + \hat{k}(B_x C_y - B_y C_x)$

$- \hat{j}(B_x C_z - B_z C_x)$

$= \hat{i}(4-9) + \hat{j}(12+4) + \hat{k}(-6-8)$

$\bar{b} \times \bar{c} = -5\hat{i} + 16\hat{j} - 14\hat{k}$ m ✓

$\bar{a} \cdot (\bar{b} \times \bar{c}) = (5\hat{i} + 4\hat{j} - 6\hat{k}) \cdot (-5\hat{i} + 16\hat{j} - 14\hat{k})$

$\bar{a} \cdot (\bar{b} \times \bar{c}) = -25 + 64 + 84 = 123 \text{ m}^3$

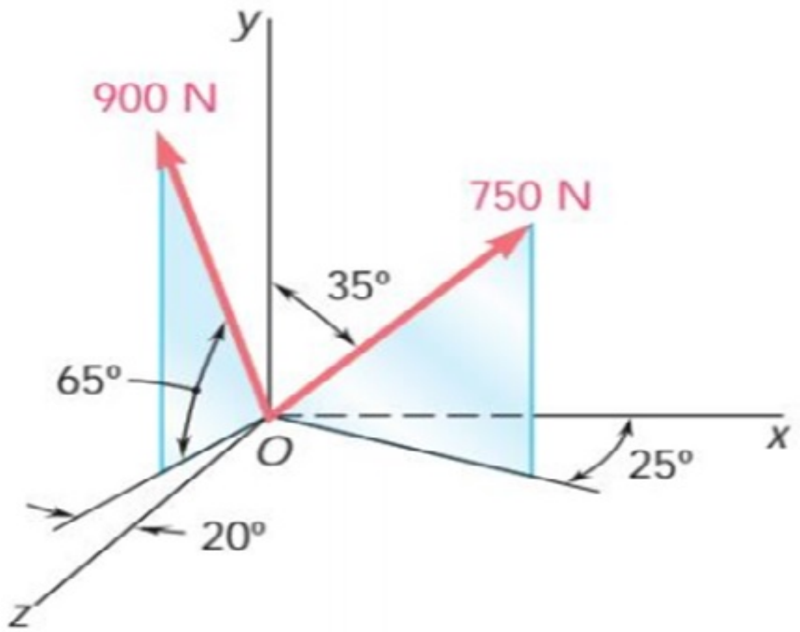
$\underbrace{\bar{b} \cdot (\bar{a} \times \bar{c}) = 123 \text{ m}^3}$

$\bar{c} \cdot (\bar{a} \times \bar{b}) = 123 \text{ m}^3$

5% f 20

100% 95

Calculate the resultant of this two vectors.



Let, $A = 900\ N$ $\quad B = 750\ N$

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$$

Now, $\boxed{B_y = 750\cos 35^\circ = 614.36}$

$$B_{zx} = 750\sin 35^\circ = 430.18$$

$$\boxed{B_x = B_{zx}\cos 25^\circ = 389.87}$$

$$\boxed{B_z = B_{zx}\sin 25^\circ = 181.80}$$

$$\vec{B} = 389.87\hat{i} + 614.36\hat{j} + 181.80\hat{k}$$

$$\boxed{A_y = 900 \sin 65^\circ = 815.67}$$

$$A_{zx} = 900 \cos 65^\circ = 380.35$$

$$\boxed{A_z = A_{zx} \cos 20^\circ = 357.41}$$

$$\boxed{A_x = -A_{zx} \sin 20^\circ = -130.08}$$

Now,

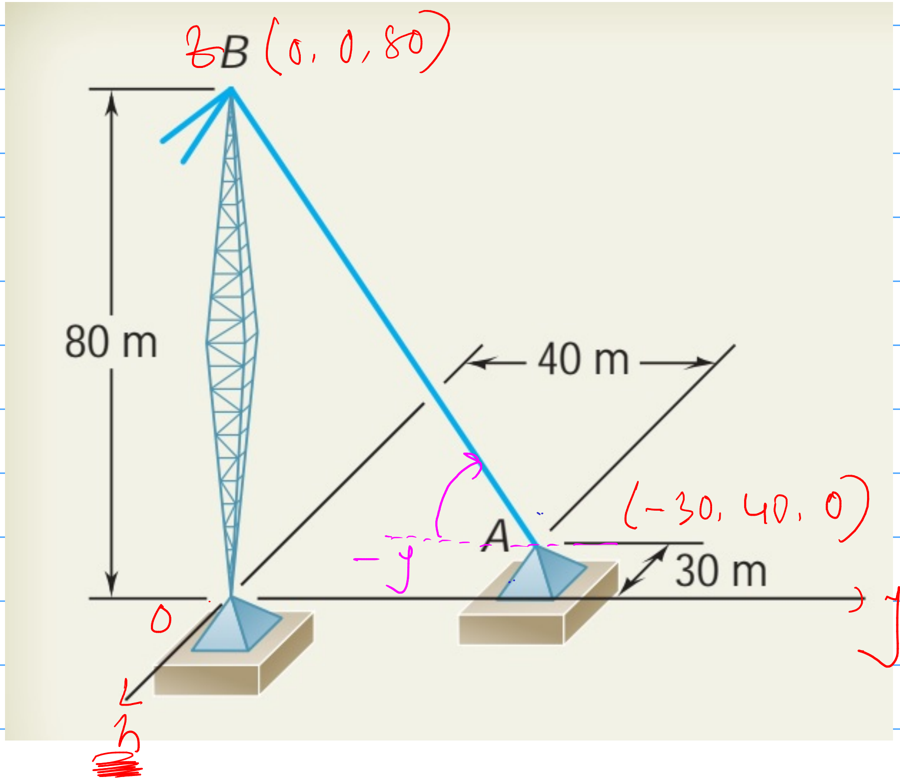
$$\boxed{\vec{A} = -130.08\hat{i} + 815.67\hat{j} + 357.4\hat{k}}$$

So,

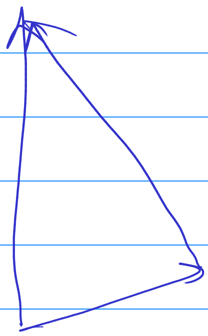
$$\boxed{\bar{A} + \bar{B} = 259.78\hat{i} + 1430.03\hat{j} + 539.21\hat{k}}$$

Calculate the length of the string.

$\vec{AB} = ?$



$\vec{OB} = 80\hat{k}$



$\vec{OA} = -30\hat{i} + 40\hat{j} + 0\hat{k}$

$\boxed{\vec{OA} = -30\hat{i} + 40\hat{j}}$

$\vec{OB} = \vec{OA} + \vec{AB}$

$\Rightarrow \vec{AB} = \vec{OB} - \vec{OA} = 80\hat{k} + 30\hat{i} - 40\hat{j}$

$\Rightarrow \vec{AB} = 30\hat{i} - 40\hat{j} + 80\hat{k}$

$\Rightarrow \boxed{|\vec{AB}| = 94.3\ m}$ Ans

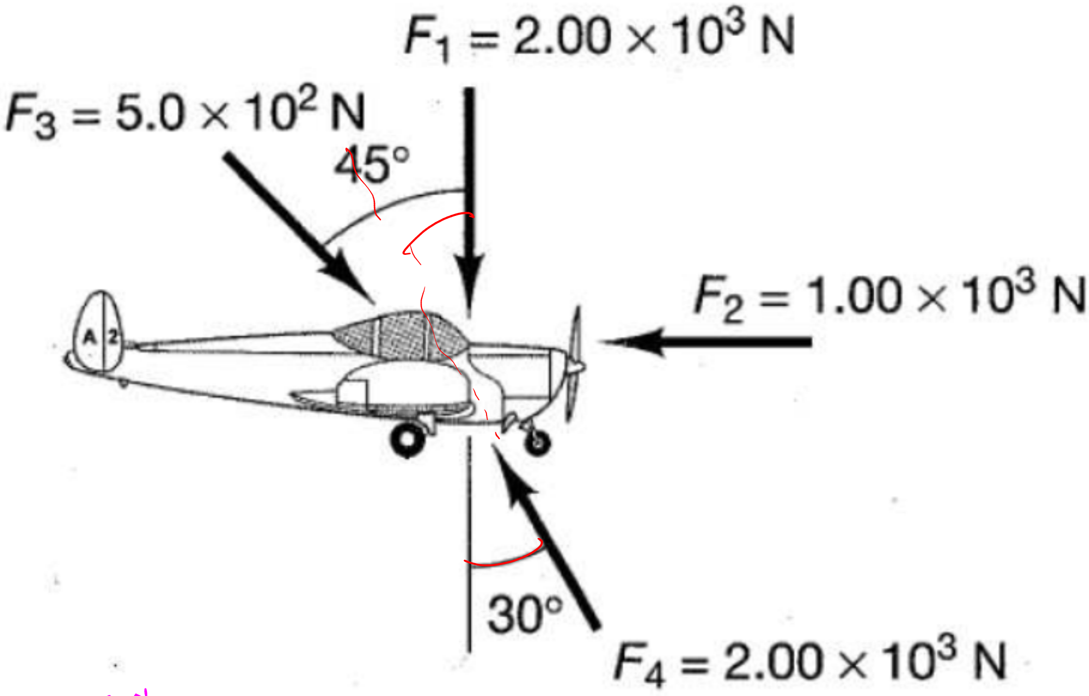
$\vec{AB}\cdot(-\hat{j}) = |AB||\hat{j}|\cos\theta$

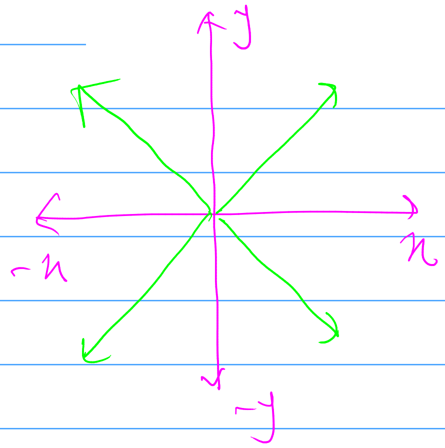
H.W.

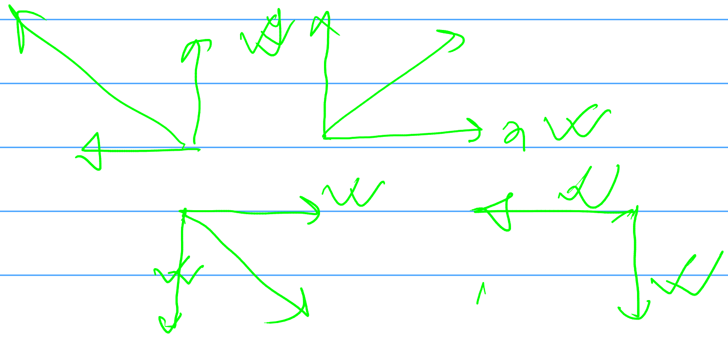
θ_x θ_y θ_z

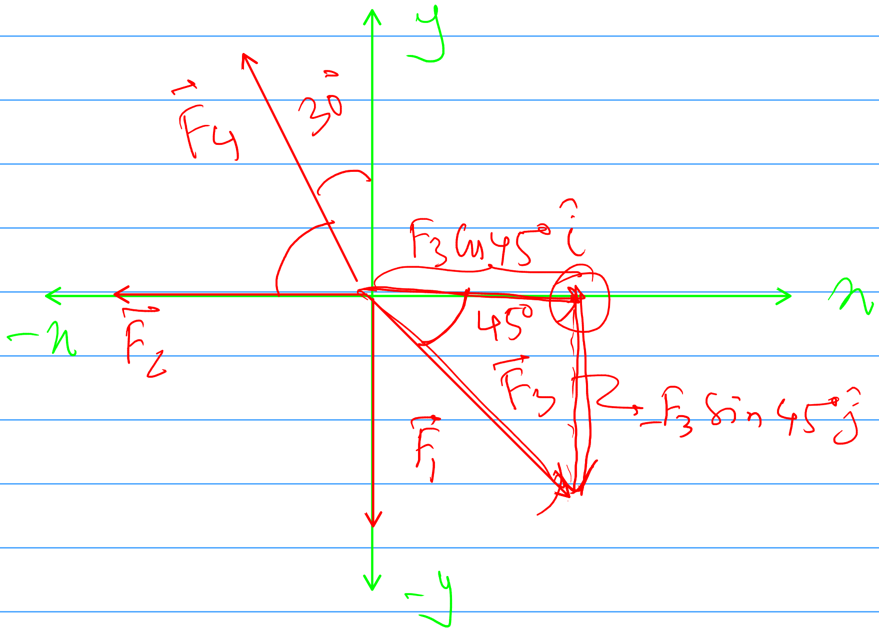
θ_{-x} θ_{-y} θ_{-z}

Calculate the resultant force F and the angle that it makes with the horizontal axis.









$$\vec{F}_1 = -F_1 \hat{j} \quad , \quad \vec{F}_2 = -F_2 \hat{i}$$

$$\vec{F}_3 = F_3 \cos 45^\circ \hat{i} - F_3 \sin 45^\circ \hat{j}$$

$$\vec{F}_4 = -F_4 \cos 60^\circ \hat{i} + F_4 \cos 30^\circ \hat{j}$$

$$\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4$$

$$\vec{F}_{net} = -F_1 \hat{j} - F_2 \hat{i} + F_3 \cos 45 \hat{i} - F_3 \sin 45^\circ \hat{j}$$
$$- F_4 \cos 60^\circ \hat{i} + F_4 \cos 30^\circ \hat{j}$$

$$\vec{F}_{net} = -1.65 \times 10^3 \hat{i} - 6.21 \times 10^2 \hat{j}$$

$$\theta = \left(\tan^{-1}\left(\frac{y}{x}\right)\right) + 180^\circ$$